METRIC

DOD-C-63982(AR)
AMENDMENT 9
23 JUNE 1989
SUPERSEDING
AMENDMENT 8
30 December 1988

MILITARY SPECIFICATION

CARTRIDGE, 30MM, HIGH EXPLOSIVE DUAL PURPOSE, M789

This Amendment forms a part of Military Specification DOD-C-63982(AR), dated 29 October 1982 and is approved for use within the US Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.2 (Drawings): Delete "11825969 - Barrel, Gun, 30mm M230 Gun" and substitute "9390748 - Barrel, Test: Gun, M230, 30mm (Progressive w/6.5° Exit Twist)". Add. "9390627 - Barrel, Test, 30mm (ADEN Barrel)".

2.1.2 (Product and Packing Drawings) - Add "12902718 - Packing and Marking of Ammunition Container, 30-mm, M592 for 30-mm Cartridges" and "12902719 - Strip, Ammunition, Metallic".

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3.8 Delete in its entirety and substitute the following:

"3.8 Muzzle velocity. The projectile shall have a mean muzzle velocity of 795-815 meters per second (m/s) at 21 degrees C ± 5 degrees C. The standard deviation shall not exceed 25 m/s at temperatures of 21 degrees C ± 5 degrees C and 71 degrees C ± 5 degrees C. The standard deviation shall not exceed 20 m/s at a temperatures of 54 degrees C ± 5 degrees C. Following the exposure to the environmental conditions of paragraph 3.10, the mean velocity at 21 degrees C ± 5 degrees C shall not increase nor decrease from the mean muzzle velocity before exposure at 21 degrees C ± 5 degrees C by more than 30 m/s.".

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3.9: Delete in its entirety and substitute the following:

"3.9 Case mouth pressure. The mean peak pressure shall not exceed 255-295 MPa at 21°C ± 5°C. The mean peak pressure shall not exceed 415 MPa at temperatures of -54°C and 71°C, respectively. Following environmental exposure, the mean peak pressure shall not increase or decrease from the mean peak pressure at 21°C ± 5°C by more than 60 MPa.".

3.11: Delete "(Dwg. 11825969)" and substitute "(Dwg. 9390627)".

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3.14.1: Delete "Barrel, Gun, 30mm - M230 (Dwg. 11825969)" and substitute "9090748 - Barrel, Test: Gun M230 (Progressive w/6.5° Exit Twist)".

* 3.14.2 Delete in its entirety and substitute the following:

"3.14.2 Static penetration. Projectiles shall be tested against 3" (76.2mm) thick rolled homogeneous armor (per MIL-A-12560G) at spin rate of 660 ± 20, 775 ± 20 and 908 ± 20 revolutions per second. The respective mean penetration depths shall exceed 1.70, 1.47 and 1.37 inches."

3.16 Should Read:

Particular care should be exercised to avoid contamination of primers and propellant by oil or grease. Such contamination may adversely affect performance. Particular care should also be exercised to avoid contamination of primers and propellant by dust and other foreign matter. If contamination by foreign matter or dust is evident, the quantity shall not exceed the requirements set forth in Drawing 11825946.

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Table I, SH 1 of 3: Add to charged Body Assembly, Proj. "Static Penetration 75 3.14/4.5.14".

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Table I:

Propellant contamination: Delete "3.17" and substitute "3.16".

Delete: "Chamber pressure" and substitute "Case mouth pressure".

Aircraft Vibration: Delete "4.5.7.2" and substitute "4.5.7.3".

Delete letter "C" under category, wherever applicable.

Delete 4.5.15 and substitute the following:

"5/ The target shall be placed 200 meters ± 20 meters from the muzzle, Use the M230 Mann Barrel."

4.4.2.1:

Major 111: Delete "3.17" and substitute "3.16".

Major 112: Delete "3.16".

Major 114: Delete "Chamber pressure" and substitute "Case mouth pressure".

Majors 117 and 118: Delete in their entirety.

Delete Major 119 in its entirety and substitute:

"Major 119 Normal Impact Function 8/, 90, Acc-2, 3.15, 4.5.15." Rej-11

Major 120: Delete "Table IV" and substitute "Table VI".

Major 122: Delete "Chamber pressure" and substitute "Case mouth pressure".

Minor 201, reference paragraph: Add "6.12".

Minor 204: Delete "3.16" and substitute "3.17"

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* In paragraph 4.2.2.2, delete Minor Examinations 204 and 205 in their entirety.

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* In paragraph 4.2.2.2, delete Minor Examinations 206, 207 and 208 in their entirety.

4.2.2.2, Minor 209: Delete "3.16" and substitute "3.17".

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4.4.2.3, Minor 202: Delete "3.16" and substitute "3.17".

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4.4.2.6:

Add "11825955" to the next higher assembly.

Delete Major 104 in its entirety and substitute Major 104: "Static penetration, 75, 100%, 3.14.2, 4.5.14.2".

Minor 201: Delete "3.16" and substitute "3.17"

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4.4.2.7: Inspection Method, Major 101 should read: Visual/Weight 1/

1/ Should Read:

"1/ Weight Method: Determine minimum flash tube/tape weight. Weigh each flash tube assembly and accept assemblies with a weight equal to or greater than the maximum pellet weight allowed plus the minimum flash tube/tape weight.".

4.4.2.7, Minor 201: Delete "3.16" and substitute "3.17".

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4.4.2.8, Minor 203: Delete "3.16" and substitute "3.16/3.17".

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4.4.2.9:

Add "11825954" to the next highest assembly number.

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Major 101: Delete "4.5.20" and substitute "4.5.16".

Major 102: Delete "3.5" and substitute "6/".

Minor 201: Delete "3.16" and substitute "3.17".

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4.4.2.10: The sum for target practice (4.4.2.4)

Delete "Critical 1" and substitute "Major 106".

Delete "in groove" and substitute "engaged".

Delete "2 boxes/lot" and substitute "5 boxes/lot 1/".

Delete all "100%" and substitute "Acc-0, Rej-1".

Minor 201: Delete "100%" and substitute ".65%" AQL.

Add to Note section: "1/ Sample size for all Majors, 5 boxes/lot".

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Note 5a: Delete "mean".

Note 5c: Delete repeated "less".

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Note 7: Delete "(see 6.12)".

Add: Footnote 8. "8/ The failure of two or less projectiles to function high order detonation shall accept the lot. If 3 - 10 failures occur, the lot will be retested per paragraph 4.5.15. Failure of 11 or more projectiles shall be cause for rejection of the lot.".

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* In "Table V. Firing Defects 1/", delete the defect criteria or "Primer" in its entirety. Substitute the following:

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"Primer	Blown primer 7/	0.25
	Loose primer 8/	0.65
	Internal minor primer leak 9/	0.65
	External minor primer leak 9/	0.65
	Major primer leak 9/	0.40
	Critical primer leak 9/	0.065"

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* Delete the paragraph "9/ Primer Leaks." in its entirety and substitute the following:

"9/ Primer Leaks. Primer leaks shall be defined as follows:

a. Suspect. A suspect primer leak is defined as the definite evidence of smudging or discoloration around any part of the periphery of the primer cup, button, insulator, or primer pocket. The fired primer must be removed by sectioning the cartridge case and must be disassembled to determine whether actual gas leakage has occured.

b. Internal minor Leak. A suspect primer which, upon disassembly shows definite evidence of a leakage path between primer components or through a crack or split in any primer component.

c. External minor primer leak. A suspect primer, which upon disassembly, shows definite evidence of a leakage path between the primer cup and primer pocket without evidence of erosion of the pocket.

d. Major leak. A Primer leak which shows erosion of any primer component.

e. Critical leak. A primer leak which shows evidence of erosion of the primer pocket or melting of any primer component.

f. A dislodged primer button is not cause for rejection.

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4.5.1: Delete "to Drawing 11825969" and substitute "to Drawing 9390627, unless otherwise specified".

4.5.3: Delete "for information only" after Data at -54°C and +71°C will be recorded.

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4.5.5: Delete "3.12" and substitute "3.13".

4.5.6: Delete "6.10" and substitute "6.13".

4.5.7.1, change the last sentence to read:

"After exposure, the test ... subjected to the tests referenced in 4.5.7."

4.5.7.2, change the last sentence to read:

"After exposure, the test ... subjected to the tests referenced in 4.5.7."

4.5.7.3:

Change the last sentence to read: "Following exposure, test item ... subjected to the tests referenced in 4.5.7."

Delete: "AAH magazine" and substitute "M592 Ammo Can".

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Figure 1: Delete in its entirety and replace with included figure (514.2-3).

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Figure 2: Delete in its entirety and replace with included figure (514.2-10).

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4.5.7.4: Delete "3.8, 3.9 and 3.12" and substitute "3.10".

4.5.9:

First sentence: Delete 3.11" and substitute "3.12".

Second sentence: Delete "3.16" and substitute "6.8".

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4.5.13:

Delete "6.19" and substitute "4.5.22".

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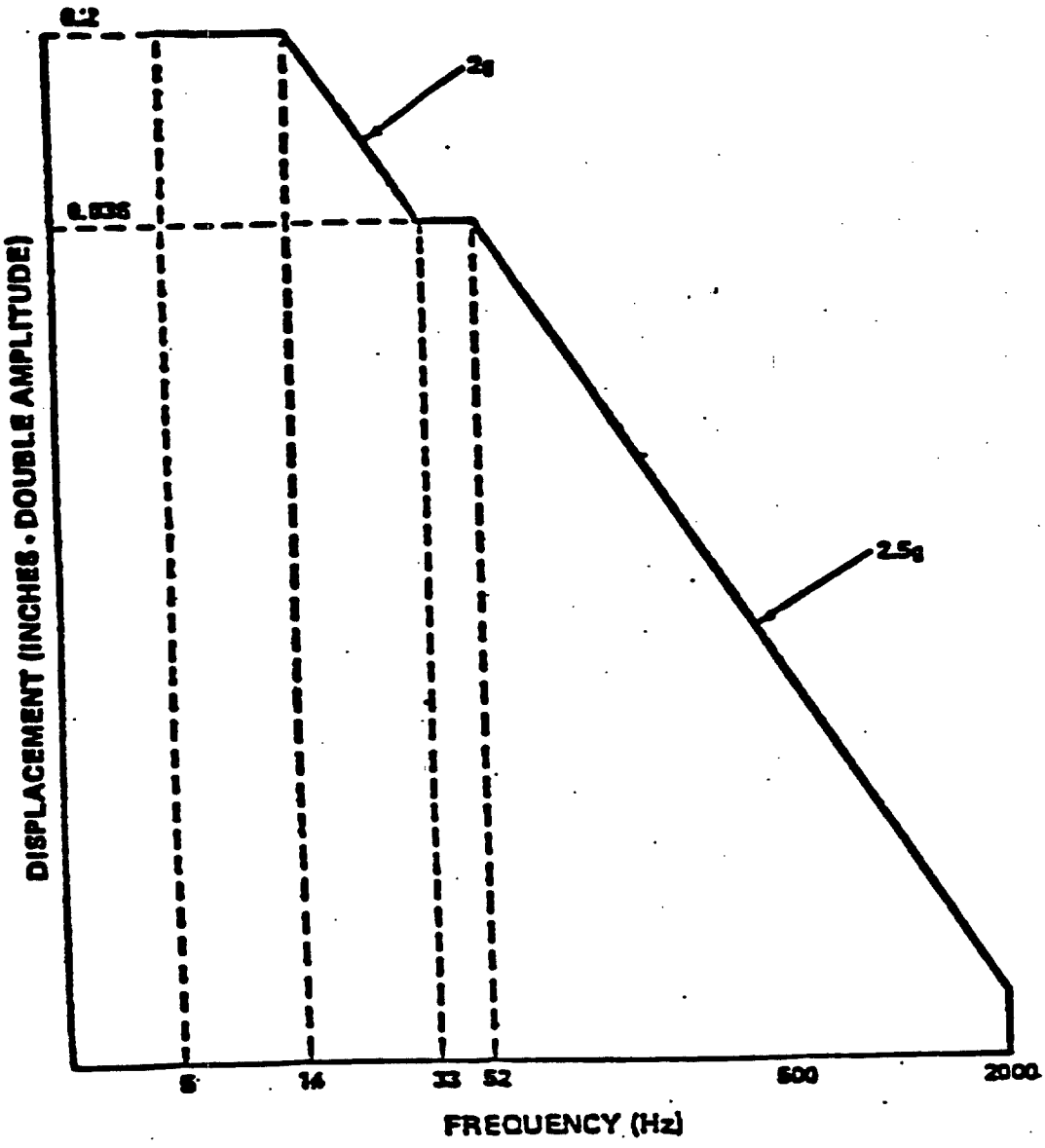


FIGURE 1. Vibration Test Curves for Equipment Installed in Helicopter. Equipment Category C. (Ref. Figure 514.2-3 of MIL-STD-810C)

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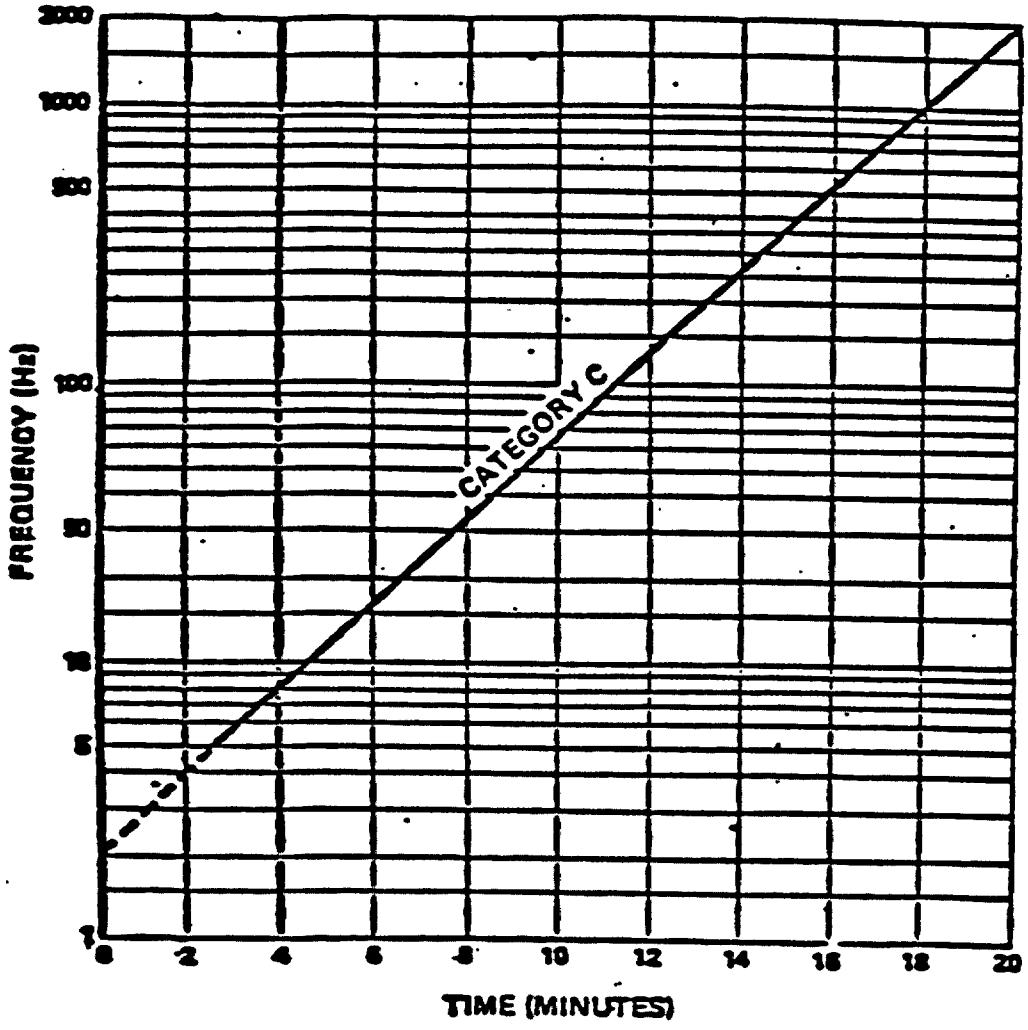


FIGURE 2. Logarithmic Sweep.
(Ref. Figure 514.2-10
of MIL-STD-810C)

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4.5.14.1: Add the following sentence "The test barrel conforming to Drawing 9390748 shall be used in this test."

4.5.15 Delete in its entirety and substitute the following:

"4.5.15 Normal impact function. The target shall be placed 200 meters ± 20 meters from the muzzle. The failure of 11 or more projectiles to function high order detonation shall be cause for rejection of the lot. If 3 - 10 failures occur, the lot will be retested with the following double sampling plan:

n_1	=90	c_1	=2	r_1	=11
n_t	=270	c_t	=10	r_{1t}	=11

If two or less failures occur during the initial test sample, the lot will be accepted. The test barrel conforming to Drawing 9390748 shall be used in this test.".

4.5.18: Add a "," between "velocity" and "pressure".

4.5.19: Delete "3.2.1.5" and substitute "3.11".

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4.5.20: Delete in its entirety

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4.5.23: Delete in its entirety.

Add paragraph 5.1.1.3 as follows:

"5.1.1.3 Bulk (Metallic Strip) packing. The Bulk (Metallic Strip) Pack shall consist of 10 30mm Metallic Ammunition Strips (see drawing 12902719) containing 11 cartridges each packed in the M592 shipping and Storage container."

Add paragraph 5.1.2.3 as follows:

"5.1.2.3 Bulk (Metallic Strip) packing. The interior packing shall be in accordance with drawing number 12902718. The 30mm rounds shall be placed in the Metallic Strips in accordance with drawing 12910289."

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Delete paragraph 5.2 in its entirety and substitute the following:

"5.2 Marking. Marking for shipment shall be in accordance with the instructions of the contract and shall comply with drawing number 11825893 (for Bulk "Metallic Strip" Packing, marking shall comply with drawing number 12902718). In addition, exterior containers shall show the caution markings required by Title 49, Code of Federal Regulations, indicating the appropriate classification of ammunition."

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6.1:

Delete "Aden MXIV" and substitute "ADEN MKIV".

Delete "gun, high explosive dual purpose M789".

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6.6: Delete "still" from last sentence.

6.7: Delete "chamber pressure" and substitute "case mouth pressure".

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6.8: Delete 4.5.12" and substitute "4.5.9".

6.12, Add d:

"d. Missing paint on the rotating band is not a defect."

PAGE 43

6.15: Delete in its entirety.

6.15 Should Read:

"6.15 Retest procedures. Failure of the material samples to comply with the velocity, pressure or action time require- ments shall result in a retest. Sample size shall be as indicated in the applicable table.".

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